MB502-Practical Exam Notes

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1. Describe in detail the major steps involved in extracting DNA from whole blood using the organic extraction method.

Answer: The major steps in organic DNA extraction from whole blood are:

1) Lysis of red blood cells using Tris EDTA buffer

2) Digestion using SDS, proteinase K buffer to lyse white blood cells

3) Phase separation using phenol-chloroform isoamyl alcohol to remove proteins

4) DNA precipitation using alcohol

5) Washing DNA pellet with ethanol

6) Diluting DNA pellet in TE buffer

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2. What is the purpose of using commercial DNA extraction kits for isolating DNA from tissues? Explain the key steps involved.

Answer: Commercial DNA extraction kits provide a rapid, convenient and efficient method to isolate high quality DNA. Key steps include: 1) Tissue disruption and homogenization

2) Incubation at 56°C with proteinase K to lyse cells

3) Column binding of DNA to selectively absorb DNA

4) Washing away proteins, lipids with alcohol based buffers

5) Eluting pure, intact genomic DNA suitable for downstream applications.

3. What is the role of gel electrophoresis in quantifying extracted DNA? Explain the methodology.

Answer: Gel electrophoresis separates DNA based on size. Comparing fluorescence intensity of DNA samples bands against known standards bands after staining with EtBr allows qualitative and semi-quantitative estimation of DNA concentration and purity. The gel is visualized under UV in a gel documentation system.

4. Explain the CTAB buffer extraction method for isolating DNA from plant tissues.

Answer: Key steps include use of CTAB buffer containing detergent and polyvinylpyrrolidone to 1) lyse plant cell wall 2) solubilize lipids and proteins 3) precipitate carbohydrates and polysaccharides. Chloroform extraction then removes proteins and phenols. RNase digestion removes RNA contaminants. DNA is finally selectively precipitated using alcohol, washed and resuspended.

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5. What is the principle of Trizol method for RNA isolation? Explain the major steps involved.

Answer: Trizol consists of phenol and guanidinium isothiocyanate which simultaneously lyses cells and inactivates endogenous RNases to maintain RNA integrity. Key steps include 1) Trizol mediated cell lysis 2) Separation of RNA containing upper aqueous phase 3) Precipitation of RNA with alcohol 4) Solubilizing RNA in RNase free water.

6. Explain the purpose and working principle of real-time PCR. What are its key components?

Answer: Real-time PCR allows amplification and simultaneous quantitative detection of DNA targets in real-time. It relies on fluorescent reporters which bind

to amplified product, emitting signals during thermocycling made possible with specialized PCR instruments. Key components are template DNA, primers, DNA polymerase, nucleotides, fluorescent probes/dyes and buffer.

7. Differentiate between VNTRs and STRs. How are these genetic markers useful in DNA fingerprinting?

Answer: VNTRs are longer 10-60bp minisatellite tandem repeats while STRs are smaller 1-6 bp microsatellite repeats but both show variability between individuals useful for forensic identification, paternity testing etc. due to the very low probability of two persons (except identical twins) showing same repeat numbers.

8. Give an overview of the procedure involved in Restriction Fragment Length Polymorphism (RFLP) analysis. What are its applications?

Answer: Key steps include extraction of DNA sample, digestion with restriction enzymes, size-separation of DNA fragments by gel electrophoresis, and analysis of banding patterns for profile matching. Applications: Determining source of DNA at crime scene, parentage verification, mapping of genes and identification of mutations.

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9. How are chemically competent E. coli cells prepared and transformed with foreign DNA? Explain.

Answer: Competent E.coli cells grown to mid-log phase are treated with divalent cations like Ca2+ (CaCl2 wash) which neutralizes surface charges, allowing DNA binding. Heat shock aids uptake of DNA which is then integrated into host genome or plasmid. Cells are then grown in selective media to identify transformants.

10. Describe the plasmid DNA isolation method from bacterial cells by alkaline lysis. How does it selectively isolate plasmid DNA?

Answer: Alkaline lysis uses SDS and NaOH to disrupt cell wall. High pH denatures chromosomal but not plasmid DNA. Upon neutralization, smaller plasmids renature correctly while denatured proteins and genomic DNA aggregate and precipitate enabling selective plasmid purification.

Here are 10 short questions with answers from the document:

1. **What are the two components needed to prepare 1M Tris-HCl buffer (pH 8)?**

Tris base 121.1g and H2O to 800ml. Adjust pH to desired value with concentrated HCl. Mix and add H2O to 1 Liter.

2. **What are the three ingredients needed to make TE buffer?**

1M Tris-HCl pH 8.0, 0.5 M EDTA pH 8.0, and H2O.

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3. **What is the role of proteinase K in DNA extraction from blood?**

Proteinase K is added to digest proteins and lyse white blood cells to release DNA during the digestion step.

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4. **What are the key steps in isolating DNA from mammalian tissue using a commercial kit?**

Key steps are tissue disruption, incubation with proteinase K, column binding of DNA, washing, and DNA elution.

5. **What dye is commonly used to visualize DNA bands in agarose gel electrophoresis?**

Ethidium bromide is commonly used to visualize DNA bands in agarose gel electrophoresis.

6. What buffer is used for DNA extraction from plant tissues?

CTAB (cetyl trimethylammonium bromide) buffer is used for DNA extraction from plant tissues.

7. What reagent is used to isolate RNA using Trizol method?

Trizol reagent, containing phenol and guanidine isothiocyanate, is used to isolate RNA.

8. How can the purity of extracted DNA/RNA be assessed?

Purity can be assessed by measuring absorbance ratio at 260/280 nm using a spectrophotometer like Nanodrop.

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9. What are the key components needed for cDNA synthesis reaction?

Key components are RNA template, reverse transcriptase, buffer, dNTPs, primers, RNase inhibitor and water.

10. What are the three main temperature steps in each PCR cycle?

The three main steps are denaturation, annealing, and extension.

Here are 10 additional short questions from the document:

1. **What are two types of DNA samples that can be purified using DNA extraction?**

Any living organism and its living parts, including human tissues, blood, hair, rodent tissues, leaf, bacteria, yeast, fungi, insect, etc.

2. **What are the main steps in organic DNA extraction method?**

Lysis of red blood cells, digestion (lysis of white blood cells), phase separation, DNA precipitation, washing, and diluting the DNA pellet.

3. **What is the purpose of using PCI in DNA extraction?**

Phenol-chloroform isoamyl alcohol (PCI) is used to remove proteinaceous materials and purify the released DNA.

4. **What is the principle behind using fluorescent dyes or probes in realtime PCR?**

As amplification progresses, the fluorescent dye or probe binds to the amplified DNA and emits fluorescence which is detected.

5. **What is the difference between VNTRs and STRs?**

VNTRs are longer (10-60 bp) minisatellite repeats while STRs are smaller (1-6 bp) microsatellite repeats.

6. **What is the first step in RFLP analysis for DNA fingerprinting?**

The first step is extraction of DNA from blood, saliva or other samples.

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7. **What is the purpose of making chemically competent E.coli cells?**

To allow efficient uptake and replication of foreign DNA like plasmids for cloning experiments.

8. **How can plasmid DNA be separated from bacterial genomic DNA?**

Alkaline lysis selectively denatures gDNA allowing smaller plasmid DNA to renature and remain in solution.

9. **Why is SDS used in SDS-PAGE protein analysis?**

SDS denatures proteins and imparts a uniform negative charge to fractionate by size during electrophoresis.

10. **What is the final step in analyzing protein bands by SDS-PAGE?**

The final step is photographing the Coomassie blue stained protein bands using a light box after destaining.

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Here are 10 more tough short questions from the document:

1. **How can contamination be prevented during preparation of competent E.coli cells for transformation?**

Strict sterile technique must be observed including use of sterile centrifuge bottles, tubes, solutions, pipets, etc.

2. **What is the purpose of adding ethidium bromide in electrophoresis agarose gel?**

Ethidium bromide intercalates with DNA and fluoresces under UV light, allowing visualization of DNA bands.

3. **What is the difference between one-step and two-step quantitative RTPCR (qRT-PCR)?**

In one-step, reverse transcription and PCR occur in one tube while two-step has separate reverse transcription and PCR.

4. **How does the principle of SYBR green dye based DNA detection differ from probe based detection in real time PCR?**

SYBR green binds any double stranded DNA while sequence specific probes bind only to complementary target sequences.

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5. **Why should monozygotic twins not have identical DNA profiles in spite of originating from the same embryo?**

Though rare, mutations can still occur independently in each twin leading to some differences in DNA profile.

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6. **What is the purpose of using alkaline phosphatase in molecular cloning experiments?**

To remove 5' phosphate groups from vector DNA ends to prevent re-ligation and improve ligation with insert DNA.

7. **How does the alkaline lysis method selectively isolate plasmid DNA from genomic DNA of bacteria?**

Alkaline conditions denature all DNA but smaller plasmid DNA renatures faster upon neutralization, trapping gDNA.

8. **What role does SDS play in protein analysis by SDS-PAGE technique?**

SDS denatures secondary/tertiary protein structures and imparts negative charge to fractionate by size only.

9. **What would happen if alcohol precipitation step is skipped while washing DNA pellet?**

Traces of salt and protein will contaminate DNA pellet leading to poor quality DNA unsuitable for downstream applications.

10. **How can non-specific DNA amplifications be identified in SYBR green real time PCR assay?**

By melt-curve analysis, non-specific products show different melting temperatures than specific product peak.

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Multiple Choice Questions

Here are 25 MCQs from the document, each with one correct option in bold:

1. What is used to isolate DNA from a wide range of biological samples?

A. CTAB buffer

B. Organic extraction

C. Alkaline lysis

D. Chromatography column

E. Commercial DNA extraction kits

2. The digestion step in organic DNA extraction involves:

A. Restriction enzymes

B. Lysozyme

C. SDS and proteinase K

D. Exonucleases

E. RNases

3. Which dye intercalates with DNA to enable visualization after gel electrophoresis?

A. Acridine orange

B. Silver nitrate

C. Coomassie blue

D. Bromophenol blue

E. Ethidium bromide

4. What makes up the two layers formed after PCI separation step?

A. DNA and lipids

B. Protein and DNA

C. Aqueous solution and phenol

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D. Organics and aqueous solution

E. Denatured proteins and nucleic acids

5. CTAB buffer is ideal for isolating DNA from:

A. Bacteria

B. Yeast

C. Human tissues D. Whole blood

E. Plant tissues

6. Which reagent is utilized in the Trizol RNA extraction method?

A. Chloroform

B. PCI

C. CTAB

D. TE buffer

E. Guanidinium thiocyanate

7. cDNA is synthesized from RNA by:

A. DNA polymerase

B. Reverse transcriptase

C. RNase H D. Primase

E. DNA ligase

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8. Real-time PCR quantification relies on:

A. Radioactive probes

B. Primers

C. Fluorescent dyes/probes D. Restriction digestion

E. DNA sequencing

9. The two genetic markers used in DNA fingerprinting are:

A. mRNA and tRNA

B. siRNA and miRNA

C. rRNA and snRNA

D. Genes and transposons

E. VNTRs and STRs

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10. What does RFLP stand for?

A. Radioactive Fragment Length Polymorphism

B. Random Fragmentation Length Polymorphism

C. Restriction Fragment Length Pattern

D. Rapid Fluorescent Length Polymorphism

E. Restriction Fragment Length Polymorphism

11. Why is CaCl2 treatment used while preparing competent cells?

A. To induce mutation

B. To allow DNA degradation

C. To enable plasmid curing

D. To induce cell lysis

E. To enable DNA binding and uptake

12. What allows selective isolation of plasmid DNA during alkaline lysis?

A. Low pH

B. SDS wash

C. Size fractionation

D. RNase treatment

E. Differential renaturation rates

13. SDS in SDS-PAGE denatures tertiary protein structures by:

A. Oxidation

B. Phosphorylation

C. Acetylation

D. Methylation

E. Disrupting hydrophobic interactions

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14. What will happen if DNA pellet is over-dried after alcohol wash?

A. Increased yield

B. Higher purity

C. Improved stability

D. Faster re-solubilization

E. Decreased solubility

15. Hot start PCR includes:

A. Low temperature annealing

B. High salt concentration

C. Initial low pH

D. Addition of DMSO

E. Delayed DNA polymerase addition

16. Why is RNase treatment used during DNA extraction?

A. To fragment RNA

B. To purify RNA

C. To stabilize RNA

D. To ligate RNA

E. To remove RNA contamination

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17. Which component controls reaction temperature in thermal cycling?

A. dNTPs

B. Taq polymerase

C. Primers

D. Template DNA

E. PCR instrument

18. What makes SYBR green DNA quantification less specific?

A. Low sensitivity

B. Small amplicon size

C. Multiple binding sites

D. Fast melting kinetics

E. Non-specificity to any double stranded DNA

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19. Why is the same DNA profile seen in monozygotic twins?

A. Due to dominance

B. Due to co-dominance

C. Due to epistasis

D. Due to linkage disequilibrium

E. Due to single embryological origin

20. What is the final step after electrophoretic separation of RFLP fragments?

A. Molecular cloning

B. Genome sequencing

C. Gene expression study

D. Mass spectrometry

E. Visualization and analysis of banding pattern

21. What is the purpose of doing a plasmid DNA extraction?

A. Gene delivery vector

B. Antibiotic resistance

C. Protein over-expression

D. Complementation assay

E. All of the above

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22. Why should teaming oxygen be added at the end while preparing acrylamide gels?

A. Increase electrophoretic rate

B. Reduce bubble formation

C. Improve polymerization

D. Prevent photo-bleaching

E. Initiate polymerization reaction

23. How can non-transformed bacterial cells be distinguished from transformants?

A. Morphology

B. Viability

C. Colony PCR

D. Growth kinetics

E. Selection by antibiotic resistance gene encoded on plasmid

24. What is the role of alkaline phosphatase in molecular cloning?

A. Cloning vector amplification

B. Insert DNA polymerization

C. Prevent self-ligation of vector

D. Ligation reaction

E. Dephosphorylation of cloning vector ends

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25. Which step helps in VNTR analysis by RFLP?

A. Cell culture

B. DNA sequencing

C. cDNA synthesis

D. DNA quanitification

E. Restriction enzyme digestion

Here are a bit more conceptual MCQs with the correct options in bold:

1. What is the full form of EDTA used in preparation of stock solutions?

A. Ethylene diamine Tetra acetic Acid

B. Ethylenediamine Tetra acetic Acid

C. Ethylenediamide Tetra acetic Acid

D. Ethylene diamide Tetra acetic Acid

2. What is the purpose of using DTT in cDNA synthesis reaction mix?

A. To act as template

B. To act as enzyme

C. As a reducing agent

D. To increase pH

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3. What is the effect of using primers with very high GC content in PCR?

A. Increased chance of primer dimer formation

B. Increased specificity

C. Faster amplification

D. Higher yield

4. SDS-PAGE technique helps separating proteins based on what physical property?

A. Shape

B. Charge

C. Size

D. Solubility

5. Why bromophenol blue dye is added in the protein sample prepared for SDSPAGE?

A. To visualize the protein bands

B. To monitor the progress of electrophoresis

C. To bind to the proteins

D. To increase density of the sample

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6. Which of the following is NOT a source that can provide DNA for extraction experiments? A. Bacterial cultures

B. Blood

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C. Hair follicles

D. Leaf

7. What is the purpose of using chaotropic salts during nucleic acid extraction?

A. To denature proteins

B. To solubilize nucleic acids

C. To precipitate lipids

D. To stabilize enzymes

8. Which method of DNA quantification relies on intercalating dyes to emit fluorescence? A. Gravimetry

B. UV-Vis spectrophotometry

C. Agarose gel electrophoresis

D. HPLC

9. What is the primary purpose of performing hot start PCR?

A. To use a higher denaturation temperature

B. To enhance specificity

C. To prevent loss of enzyme activity at high initial temperatures

D. To increase the annealing temperature

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10. Which type of DNA fingerprinting method has now been rendered almost obsolete?

A. PCR based

B. RFLP based

C. DNA sequencing based

D. Southern blotting based

11. What is the main utility of performing cDNA synthesis in an experiment?

A. Storing RNA for future use

B. Converting RNA to more stable DNA form

C. Increasing abundance of RNA

D. Separating RNA fragments

12. Which enzyme is added during the neutralization step in alkaline lysis method of plasmid DNA extraction?

A. Restriction enzyme

B. DNA ligase

C. Reverse transcriptase

D. RNase

13. What is the purpose of adding SDS during bacterial cell lysis for plasmid DNA extraction?

A. To aid in cell wall disruption

B. To facilitate plasmid renaturation

C. To completely denature DNA

D. To precipitate chromosomal DNA

14. What is the source of the RNase A enzyme added during CTAB extraction of plant DNA? A. Calf intestine

B. Bacteria

C. Yeast

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D. Bovine pancreas

15. Which dye is most commonly used for DNA quantification in agarose gel electrophoresis? A. Acridine orange

B. Ethidium bromide

C. Crystal violet

D. Methylene blue

16. What is the main difference between cDNA and gDNA in terms of origin?

A. cDNA comes from RNA while gDNA comes directly from cell's DNA

B. cDNA contains only coding regions while gDNA contains introns too

C. cDNA is single stranded while gDNA is double stranded

D. cDNA contains UTRs while gDNA does not

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17. What is the purpose of adding yeast extract and tryptone in LB medium for E. coli growth?

A. To increase broth viscosity

B. As sources of nitrogen and vitamins

C. To adjust pH

D. To enhance plasmid yield

18. Why is the use of sterile technique critical while preparing chemically competent cells?

A. To prevent cell clumping

B. To enhance transformation efficiency

C. To prevent contamination

D. To improve cell survival

19. Which of the following is NOT required for visualizing DNA bands in an agarose gel?

A. Electrophoresis apparatus

B. UV light source

C. Camera

D. Intercalating dye

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20. What is the main utility of alkaline phosphatase in cloning workflows?

A. Ligating DNA fragments

B. Preventing self-ligation of vectors

C. DNA amplification

D. Fragmentation of DNA

21. Why is the use of RNase-free water essential in protocols involving RNA?

A. To prevent RNA hydrolysis

B. To enhance enzyme activity

C. To improve yield

D. To remove genomic DNA

22. Which of the following components is NOT required in a PCR reaction mix?

A. Template DNA

B. Taq polymerase

C. RNase inhibitor

D. Primers

23. Why is the cDNA synthesized in two-step RT-PCR transferred to a separate tube before performing PCR?

A. To change the buffer composition

B. To dilute out inhibitors

C. To enhance quality

D. To avoid loss of reagents

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24. What is the purpose of adding gel loading dye to DNA samples before running agarose gel electrophoresis?

A. To stain the DNA

B. To increase density for easy loading

C. To visualize DNA migration

D. To quantify DNA

25. Which method of DNA extraction involves use of chaotropic salts and silica columns?

A. Phenol-chloroform

B. CTAB

C. Spin column kits

D. Organic solvents Here are the correct options for the multiple choice questions in full form:

1. Ethylenediaminetetraacetic Acid
2. As a reducing agent
3. Increased chance of primer dimer formation
4. Size
5. To monitor the progress of electrophoresis
6. Leaf
7. To solubilize nucleic acids
8. UV-Vis spectrophotometry
9. To prevent loss of enzyme activity at high initial temperatures

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10. Southern blotting based

11. Converting RNA to more stable DNA form

12. RNase

13. To facilitate plasmid renaturation

14. Bovine pancreas

15. Ethidium bromide

16. cDNA comes from RNA while gDNA comes directly from cell's DNA

17. As sources of nitrogen and vitamins

18. To prevent contamination

19. Camera

20. Preventing self-ligation of vectors

21. To prevent RNA hydrolysis

22. RNase inhibitor

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23. To dilute out inhibitors

24. To increase density for easy loading

25. Spin column kits

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